Week 6

Case Study 1

Data for Questions 1 to 5

Machine Reading Compression

A maintenance team receives pairs of machine readings from an industrial unit and wants a compact summary that can still reconstruct the original reading reasonably well. To understand the mechanics of an autoencoder, the team studies one two-feature reading $x = (4, 2)^\top$ and compresses it into one latent value using an encoder

$$z = 0.5x_1 + 0.5x_2.$$

The decoder reconstructs

$$\hat{x}_1 = z, \qquad \hat{x}_2 = z.$$

The reconstruction loss is

$$L = (x_1 - \hat{x}_1)^2 + (x_2 - \hat{x}_2)^2.$$

This example lets the team separate the encoder, decoder, and reconstruction-loss roles before moving to higher-dimensional machine logs.

Question 1: What is the latent value z?

A) 3

B) 2

C) 4

D) 1

Correct Answer: A

Explanation: $z = 0.5(4) + 0.5(2) = 3$.

Question 2: What is the reconstructed vector $\hat{x}$?

A) $(2, 2)^\top$

B) $(3, 3)^\top$

C) $(4, 4)^\top$

D) $(3, 2)^\top$

Correct Answer: B

Explanation: The decoder sets both reconstructed coordinates equal to $z = 3$.

Question 3: What is the reconstruction loss?

A) 4

B) 1

C) 2

D) 0

Correct Answer: C

Explanation: $L = (4-3)^2 + (2-3)^2 = 1 + 1 = 2$.

Question 4: Which part of the autoencoder creates the latent representation?

A) The decoder output layer

B) The reconstruction loss

C) The target label table

D) The encoder layer

Correct Answer: D

Explanation: The encoder maps the input x into the latent value z.

Question 5: What is the training objective of this autoencoder?

A) To minimize reconstruction error.

B) To assign external class labels before any reconstruction is attempted.

C) To ignore the input features.

D) To make every latent value zero.

Correct Answer: A

Explanation: Autoencoders are trained to reconstruct their inputs with small error.

Case Study 2

Data for Questions 6 to 10

Report Feature Compression

A document system summarizes operational reports using five numerical features such as length, keyword density, and section-level counts. The team wants a compact representation that captures the main structure without storing all five features directly. It uses 5 input features and compresses them into a latent representation with 2 hidden units before reconstructing the 5 features. The model is undercomplete because the latent dimension is smaller than the input dimension. The setup is used to discuss why compression can force the model to learn useful structure instead of simply copying every input coordinate.

Question 6: What is the input dimension?

A) 2

B) 5

C) 7

D) 10

Correct Answer: B

Explanation: The case study states that there are 5 input features.

Question 7: What is the latent dimension?

A) 1

B) 5

C) 2

D) 7

Correct Answer: C

Explanation: The latent layer has 2 hidden units.

Question 8: Why is the model undercomplete?

A) The decoder has no reconstruction target.

B) The output layer ignores all reconstructed features during training.

C) The input has fewer features than the latent layer.

D) The latent dimension is smaller than the input dimension.

Correct Answer: D

Explanation: An undercomplete autoencoder compresses inputs into a lower-dimensional latent representation.

Question 9: Which output dimension is required to reconstruct the original input?

A) 5

B) 1

C) 2

D) 3

Correct Answer: A

Explanation: The reconstruction must have the same dimension as the 5-feature input.

Question 10: How is this related to PCA in the linear case?

A) It must always produce a one-hot class label.

B) It can learn a low-dimensional representation similar to PCA.

C) It removes the need for reconstruction in every training example.

D) It makes every component have zero variance.

Correct Answer: B

Explanation: A linear undercomplete autoencoder can learn a subspace related to PCA.

Case Study 3

Data for Questions 11 to 15

Meter Reading Recovery

A utility company receives meter readings that sometimes lose one coordinate because of temporary communication noise. To make the representation robust, it trains an autoencoder using corrupted inputs but clean targets. For one example, the clean reading is

$$x = (5, 1)^\top,$$

and the corrupted input is

$$\tilde{x} = (5, 0)^\top.$$

The model outputs $\hat{x} = (4, 1)^\top$, and the loss is

$$L = (x_1 - \hat{x}_1)^2 + (x_2 - \hat{x}_2)^2.$$

During training, the encoder sees the corrupted vector, but the loss is measured against the clean reading so the model learns to recover stable structure.

Question 11: Which vector is given to the encoder during training?

A) $x = (5, 1)^\top$

B) $\hat{x} = (4, 1)^\top$

C) $(0, 0)^\top$

D) $\tilde{x} = (5, 0)^\top$

Correct Answer: D

Explanation: A denoising autoencoder receives the corrupted input $\tilde{x}$.

Question 12: Which vector is the clean reconstruction target?

A) $(4, 1)^\top$

B) $(5, 0)^\top$

C) $(5, 1)^\top$

D) $(0, 1)^\top$

Correct Answer: C

Explanation: The clean target is $x = (5, 1)^\top$.

Question 13: What is the reconstruction loss?

A) 0

B) 2

C) 4

D) 1

Correct Answer: D

Explanation: $L = (5-4)^2 + (1-1)^2 = 1$.

Question 14: What is the purpose of using corrupted inputs?

A) To train the model to recover clean structure.

B) To remove the decoder from the network before training begins.

C) To force every output to equal zero.

D) To avoid computing any reconstruction loss.

Correct Answer: A

Explanation: Denoising autoencoders learn to reconstruct clean inputs from corrupted versions.

Question 15: Which statement best describes this training setup?

A) The model ignores all input values.

B) The model maps noisy input toward a clean target.

C) The target is the corrupted vector only.

D) The encoder output is used as the final label for every noisy example.

Correct Answer: B

Explanation: The corrupted vector is input, while the clean vector is the target.

Case Study 4

Data for Questions 16 to 20

Compact Representation Control

A research group trains an autoencoder for compact feature learning, but plain reconstruction alone produces representations that are too active and sensitive for the intended analysis. To shape the latent representation, the group starts with reconstruction loss $L_{\text{rec}} = 0.8$. It adds a sparsity penalty of 0.3 with weight $\lambda = 2$, so

$$L_{\text{sparse}} = L_{\text{rec}} + \lambda(0.3).$$

In a second experiment, the group uses a contractive penalty to make the representation less sensitive to small input changes. The two penalties illustrate different regularization goals: sparsity encourages only a few hidden units to respond strongly, while contractive regularization encourages stability.

Question 16: What is the sparsity penalty contribution?

A) 0.3

B) 0.8

C) 0.6

D) 1.1

Correct Answer: C

Explanation: The weighted penalty is $\lambda(0.3) = 2(0.3) = 0.6$.

Question 17: What is L_{sparse}?

A) 1.4

B) 1.1

C) 0.8

D) 0.6

Correct Answer: A

Explanation: $L_{\text{sparse}} = 0.8 + 0.6 = 1.4$.

Question 18: What does sparse regularization encourage?

A) Every hidden unit must always be active during the full reconstruction process.

B) The decoder is removed from the model.

C) Only a small number of hidden units should be active.

D) The input dimension must equal zero.

Correct Answer: C

Explanation: Sparse autoencoders encourage most hidden activations to be small or inactive.

Question 19: What is the goal of a contractive penalty?

A) To force reconstruction loss to be ignored throughout all training.

B) To make all weights increase without bound after every input change.

C) To replace the encoder with class labels.

D) To make the representation less sensitive to small input changes.

Correct Answer: D

Explanation: Contractive penalties encourage stable representations under small input perturbations.

Question 20: Why are such penalties added to the reconstruction loss?

A) To remove all training data from the objective.

B) To control the learned representation beyond plain reconstruction.

C) To stop the model from producing reconstructions.

D) To make every feature identical to every other feature before encoding.

Correct Answer: B

Explanation: Regularization shapes what the autoencoder learns in addition to minimizing reconstruction error.